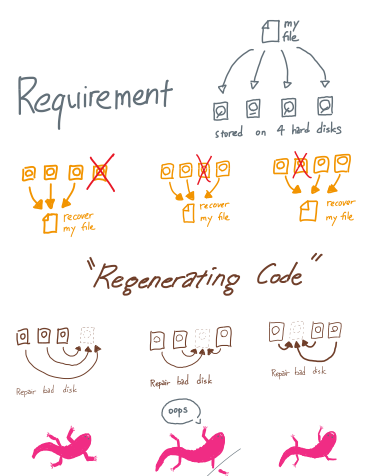


How To Speak Tensors

(with 4 interesting examples)

Hsin-Po Wang (National Taiwan University)

Cloud Storage



Use Tensor

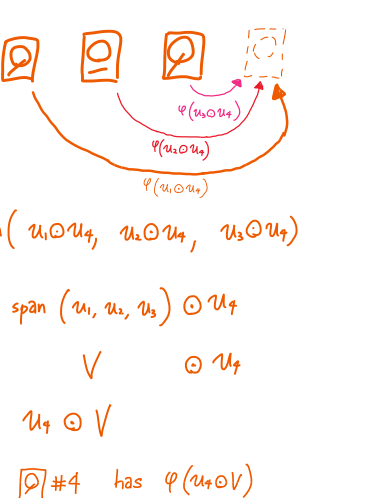
Pick a finite field $\mathbb{F}$, $k \geq 4$
Let $V = \mathbb{F}^3$
Encode my file as a map
 $\varphi: S^3 V \rightarrow \mathbb{F}$
where $S^3 V := V \otimes V \otimes V / \langle \sum_{i,j,k} a_{ijk} v_i \otimes v_j \otimes v_k \rangle$
Elements of $S^3 V$ are linear combinations of a_{ijk}

Storage: $\otimes$ #1 stores $\varphi(u_1 \otimes V)$
 $\otimes$ #2 stores $\varphi(u_2 \otimes V)$
 $\otimes$ #3 stores $\varphi(u_3 \otimes V)$
 $\otimes$ #4 stores $\varphi(u_4 \otimes V)$

File recovery: If $\otimes$ #4

$u_1 \otimes V + u_2 \otimes V + u_3 \otimes V$
 $= \text{span}(u_1, u_2, u_3) \otimes V$
 $= V \otimes V$
 $= S^2 V$
 $\Rightarrow$ we have full φ

Disk Reiparment



Matrix Multiplication

Want to compute

$$\begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

Addition & subtraction is free.

How many multiplications do we need?

Answer: 8, because

$$\begin{aligned} C_{11} &= a_{11} \cdot b_{11} + a_{12} \cdot b_{21} \\ C_{12} &= a_{11} \cdot b_{12} + a_{12} \cdot b_{22} \\ C_{21} &= a_{21} \cdot b_{11} + a_{22} \cdot b_{21} \\ C_{22} &= a_{21} \cdot b_{12} + a_{22} \cdot b_{22} \end{aligned}$$

But actually 7 is enough

$$\begin{aligned} M_1 &= (a_{11} + a_{22}) \cdot (b_{11} + b_{22}) \\ M_2 &= (a_{21} + a_{22}) \cdot b_{11} \\ M_3 &= a_{11} \cdot (b_{22} - b_{21}) \\ M_4 &= a_{22} \cdot (b_{21} - b_{11}) \\ M_5 &= (a_{11} + a_{12}) \cdot b_{22} \\ M_6 &= (a_{21} - a_{11}) \cdot (b_{11} + b_{12}) \\ M_7 &= (a_{22} - a_{12}) \cdot (b_{21} + b_{22}) \end{aligned}$$

Because

$$\begin{aligned} C_{11} &= M_1 + M_4 - M_5 + M_7 \\ C_{12} &= M_3 + M_5 \\ C_{21} &= M_2 + M_4 \\ C_{22} &= M_1 - M_2 + M_3 + M_6 \end{aligned}$$

"Strassen Alg"

Define $\langle n, n, n \rangle :=$
the cost of multiplying
 $n \times n$ with $n \times n$

Question: find ω s.t.
 $\langle n, n, n \rangle \approx n^\omega$

Exercise: $\langle 5, 5, 5 \rangle \leq$

$$\begin{aligned} &\langle 2, 2, 2 \rangle + \langle 2, 3, 2 \rangle + \langle 2, 3, 3 \rangle + \langle 2, 3, 3 \rangle + \\ &\langle 2, 3, 2 \rangle + \langle 2, 3, 2 \rangle + \langle 2, 3, 3 \rangle + \langle 2, 3, 3 \rangle \\ &= \langle 2, 2, 2 \rangle + 3 \langle 2, 3, 3 \rangle + 3 \langle 2, 3, 3 \rangle + \langle 2, 3, 3 \rangle \end{aligned}$$

Hint: $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

Lesson: knowing small $\langle n, n, n \rangle$
help us bound ω because
 $\langle n, n, n \rangle \leq \langle n, n, n \rangle + \langle n, n, n \rangle + \langle n, n, n \rangle$
 $= \langle n, n, n \rangle$

What if we know $\langle p, q, r \rangle + \langle a, b, c \rangle$?

Idea: $\left([p, q, r] \otimes [q, r, p] \otimes [r, p, q] \otimes [a, b, c] \otimes [b, c, a] \otimes [c, a, b] \right)^{\otimes m}$

$$= \bigoplus_{\substack{m_1, \dots, m_6 \\ m_1 + \dots + m_6 = m}} \binom{m}{m_1, \dots, m_6} [p, q, r]^{\otimes m_1} \otimes [q, r, p]^{\otimes m_2} \otimes \dots \otimes [c, a, b]^{\otimes m_6}$$

Quantum & Testability

Bloch Sphere

describes the spin of an electron.

Spin up $\uparrow$ Spin down $\downarrow$ Spin plus $\rightarrow$ Spin minus $\leftarrow$
 $| \uparrow \rangle$ $| \downarrow \rangle$ $| + \rangle$ $| - \rangle$

We store quantum information using spins

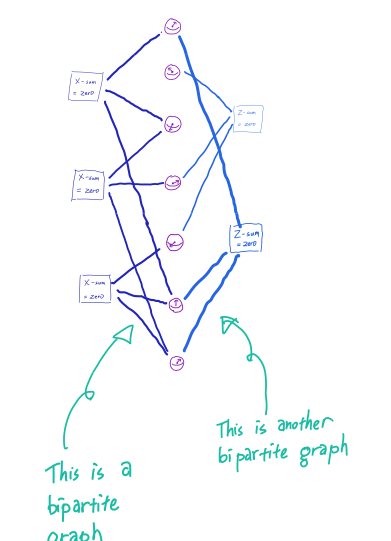
But there could be two types of errors

X-error = $G \cdot 180^\circ$ Z-error = $C \cdot 180^\circ$

A quantum fact: other errors

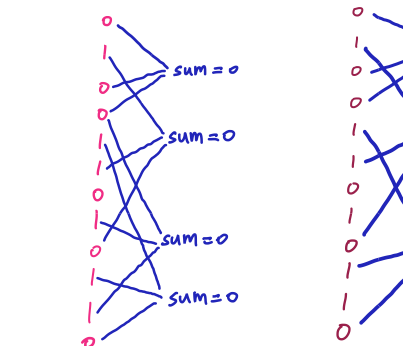
are just combination of X & Z-errors

To protect quantum information,
use X-checks & Z-checks



bipartite graph $\Rightarrow$ adjacency matrix $\Rightarrow$ linear map

This is analogous to
classical parity-check codes



syndromes are redundant

syndromes

symbols

$$\mathbb{F}_2^a \xrightarrow{f_X} \mathbb{F}_2^n \xrightarrow{f_Z} \mathbb{F}_2^b \quad (*)$$

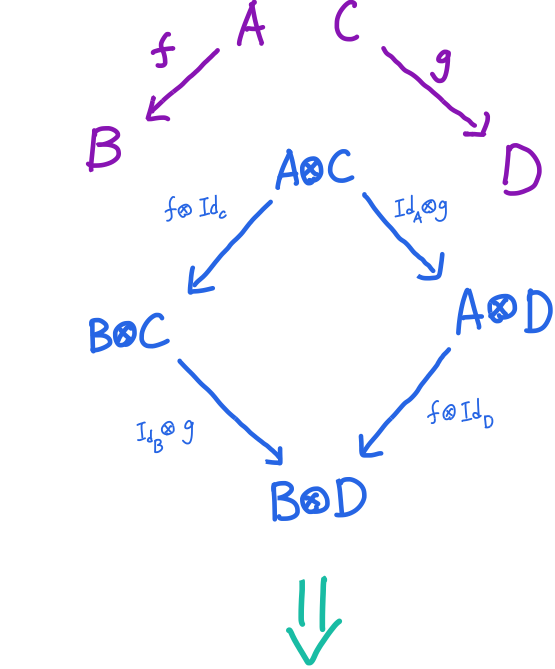
A quantum fact: we need $f_Z \circ f_X = 0$

So (*) is a chain complex

How to construct chain complex?

Hastings-Haah-Donnell:

use tensor product



$$B \otimes C \longrightarrow A \otimes C \oplus B \otimes D \longrightarrow A \otimes D$$

5. arXiv:2402.07476 [pdf, other] quant-ph cs.CG cs.IT
Expansion of higher-dimensional cubical complexes with application to quantum locally testable codes
Authors: Irit Dinur, Ting-Chun Lin, Thomas Vidick
Abstract: We introduce a high-dimensional cubical complex, for any dimension $t \geq 0$, and apply it to the design of quantum locally testable codes. Our complex is a natural generalization of the constructions by Panteleev and Kalachev and by Dinur et. al of a square complex (case $t=2$), which have been applied to the design of classical locally testable codes (LTC) and quantum low-density parity check codes (qLDPC) respectively. We turn the geometric (cubical) complex into a chain complex by relying on constant-sized local codes $h_1, \dots, h_t$ as gadgets. A recent result of Panteleev and Kalachev on existence of tuples of codes that are product expanding enables us to prove lower bounds on the cycle and co-cycle expansion of our chain complex. For $t=4$ our construction gives a new family of "almost-good" quantum LTCs -- with constant relative rate, inverse-polylogarithmic relative distance and soundness, and constant-size parity checks. Both the distance of the quantum code and its local testability are proven directly from the cycle and

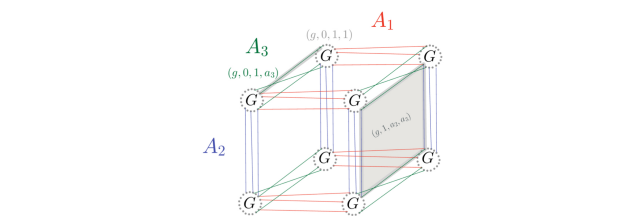


Figure 1: A cubical complex with $t = 3$ and three sets of permutations A_1, A_2, A_3 . The underlying graph can be seen to be $2^t = 8$ -partite. We label the faces as $\{f_0, f_1, f_2, f_3\}$.

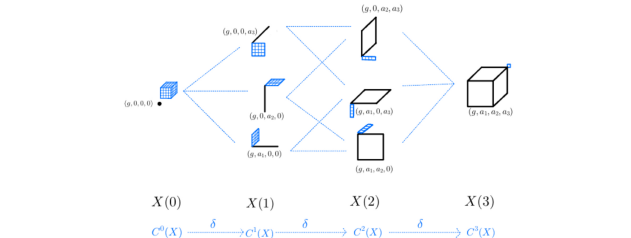


Figure 2: The co-chain complex, localized at a vertex $f = (g, 0, 0, 0)$; with $t = 3$. The geometric elements of $X_{2,f}$ are drawn in black. The associated coefficient spaces are drawn in blue.

[10] arXiv:2006.10086 [pdf, ps, other]
Multilinear Algebra for Minimum Storage Regenerating Codes
Iwan Duursma, Hsin-Po Wang
Comments: 36 pages, 9 figures, 3 tables
Subjects: Information Theory (cs.IT); Commutative Algebra (math.AC)

[11] arXiv:2006.08911 [pdf, ps, other]
Multilinear Algebra for Distributed Storage
Iwan Duursma, Xipeng Li, Hsin-Po Wang
Comments: 33 pages, 8 figures, 1 table
Subjects: Information Theory (cs.IT); Commutative Algebra (math.AC)

FIGURE 2. Step by step downloading scheme. Step 1 is the leaf check. Step $4j+2$ is restricting from nodes. Step $4j+3$ is (S_k) . Step $4j+4$ is restricting. Step $4j+5$ is induction hypothesis and coevolve-multiplication.

[14] arXiv:2011.15082 [pdf, other]
Parity-Checked Strassen Algorithm
Hsin-Po Wang, Iwan Duursma
Comments: 40 pages, 18 figures, 2 tables; v2 adds MSC and applies corrections; v3 applies corrections
Subjects: Information Theory (cs.IT); Combinatorics (math.CO)

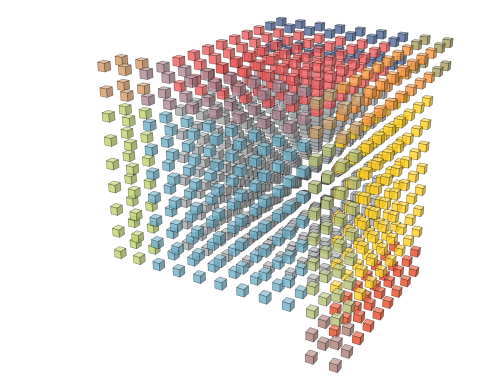


FIGURE 8. The union of four strategies— $9 \cdot 9 \cdot 9$ and $53 \cdot 9$ along with $9 \cdot 9 \cdot 53$ together with $\rho(9 \cdot 53)$. Slices in every direction fall back to figure 5 union figure 7, or equivalently, $9 \cdot 9$ union 53.

Law of large number
 $\approx \left(\frac{m}{6} \frac{m}{6} \frac{m}{6} \frac{m}{6} \frac{m}{6} \frac{m}{6} \right) [pqrabc, pqrabc, pqrabc]^{\otimes \frac{m}{6}}$
Corollary: $(pqr)^{\omega} + (abc)^{\omega} \leq \langle p, q, r \rangle + \langle a, b, c \rangle$